

Nikhilesh Vijay Chirutala

Full-Stack Software Engineer

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SUMMARY

Full-Stack Software Engineer with 2.8+ years of experience developing and supporting production-grade applications across AI, CRM, trading, and cloud contact center platforms. Strong hands-on expertise in Python, FastAPI, JavaScript, React.js, Next.js, REST APIs, MongoDB, AWS, and cloud application development. Experienced in building backend services, responsive interfaces, AI/LLM integrations, RAG-based applications, authorization mechanisms, third-party integrations, production automation, troubleshooting, testing, and deployment workflows. Proven ability to contribute across backend, frontend, cloud, AI, testing, deployment, and production support while delivering dependable and scalable software solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap, Jinja2

Backend: FastAPI, Node.js, Express.js, REST APIs, API Development, Backend Automation, Authentication, RBAC, ACL

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Databases: MongoDB, Neon DB, SQL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

APIs & Integrations: Twilio, GitHub API, REST APIs, Postman

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

- Building Python-based backend services and automation for the Convoso cloud contact center platform, handling call recordings, cloud storage, metadata synchronization, and recovery workflows for 10,000+ recordings each month.
- Engineered a call-recording recovery and S3 storage management solution from the ground up, introducing automation where manual processes previously existed and improving the reliability of production recording operations.
- Created automated workflows to detect, recover, organize, and synchronize recording assets across storage systems while preserving related metadata and application-to-cloud data consistency.
- Leveraged Python and AWS services to implement dependable cloud workflows covering recording operations, data processing, storage management, and ongoing production support.
- Diagnosed production failures across the recording lifecycle by tracing issues through backend services, storage operations, metadata synchronization, and deployment environments.
- Executed deployment verification, functional testing, defect resolution, feature improvements, and release validation to maintain stability of backend changes across development and production.
- Partnered with engineering and QA teams to translate operational requirements into fixes and enhancements, investigate root causes, and improve recording recovery and storage processes.

Key Projects:

CONVOSO — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, REST APIs

- Built and enhanced Python backend services for the Convoso cloud contact center platform, covering lead routing, call recording operations, data processing, cloud storage, and automated operational workflows.
- Engineered recording-recovery workflows capable of handling 10,000+ recordings monthly and restoring missing, unavailable, or inconsistent recording data.
- Implemented Amazon S3 storage workflows to manage recording objects, verify storage states, and keep cloud files aligned with application-level metadata.
- Developed metadata synchronization mechanisms that maintained accurate recording information between application databases, services, and cloud storage.
- Automated recurring recording-management and recovery activities, minimizing manual operational effort and improving workflow consistency.
- Added validation, exception handling, and failure-management logic to identify unsuccessful recording operations, handle unexpected storage conditions, and strengthen recovery reliability.
- Integrated REST APIs and backend components to connect recording workflows with existing contact center services and downstream data-processing pipelines.
- Analyzed production behavior, backend execution paths, storage states, and data mismatches to isolate root causes and deliver corrective changes.
- Conducted functional testing and deployment checks for backend enhancements, validating recording recovery and storage behavior across development and production environments.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Contributed as a full-stack software engineer across four production applications, covering solution design, feature implementation, backend development, frontend delivery, testing, deployment, debugging, and application support.
- Architected and implemented reusable backend modules with Python, FastAPI, REST APIs, and MongoDB, translating business requirements into scalable application services and workflows.
- Created secure REST endpoints with authentication, role-based permissions, and ACL authorization to control access across users, teams, application resources, and operations.
- Delivered frontend features with React.js, Next.js, Angular, JavaScript, TypeScript, HTML, and CSS while connecting interactive interfaces with backend APIs and application services.
- Connected external platforms and APIs such as Twilio and GitHub API to extend product capabilities, integrate communication workflows, and automate application processes.
- Supported AWS and Vercel deployments, maintained GitHub Actions CI/CD pipelines, and worked across Linux-based development and deployment environments.

Key Projects:

AI Bull — AI-Powered Trading Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment analysis platform with capabilities for stock and mutual fund analysis, screening, strategy evaluation, and historical back-testing.
- Designed and implemented a Retrieval-Augmented Generation (RAG) based chatbot that retrieves relevant financial and application-specific context before generating responses to user questions.
- Built the RAG retrieval workflow to connect user queries with relevant knowledge sources, improving contextual grounding of LLM-generated responses within the trading application.
- Integrated vector-based retrieval concepts to identify relevant information for conversational queries and provide the LLM with targeted context instead of relying only on model knowledge.
- Applied prompt engineering techniques to structure retrieved context and guide the LLM toward relevant, coherent, and application-aware answers.
- Connected the RAG chatbot with backend APIs and application workflows so users could access relevant financial information through a conversational interface.
- Developed FastAPI and Python backend services responsible for financial data processing, business logic, chatbot workflows, and supporting application operations.
- Implemented back-testing functionality that enabled users to evaluate investment strategies against historical market data and compare potential strategy outcomes.
- Created configurable screening and filtering functionality for stocks and mutual funds using user-defined financial criteria.
- Built React.js and Next.js frontend features that connected interactive financial analysis capabilities with backend APIs and AI-driven application workflows.

CRM.io — AI-Driven CRM System

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs

- Implemented core CRM functionality for managing contacts, tasks, communication activities, and telephony-related operations.
- Developed FastAPI and Python services containing application logic, REST endpoints, user workflows, and data-management operations.
- Applied role-based access control and ACL authorization to protect CRM resources and enforce permissions according to user roles and access levels.
- Integrated Twilio APIs to provide telephony and communication capabilities within the CRM application and its workflows.
- Created server-rendered interfaces with Jinja2 and connected user-facing workflows with backend services and APIs.
- Validated APIs and end-to-end application workflows using Postman while contributing to debugging, defect resolution, and production support activities.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Built a browser-based JavaScript playground where users can write, execute, test, and experiment with JavaScript through an interactive web interface.
- Connected GitHub APIs to enable project and repository-related functionality and streamline associated workflows.
- Developed and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Created a project management platform for structuring projects, tasks, collaboration activities, and team workflows.
- Implemented authentication and user-management capabilities using Clerk for secure application access.
- Integrated Neon DB for persistent data storage and deployed the application through Vercel.

Guider — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Developed an interactive learning portal with Angular 19 and TypeScript using server-side rendering for the application frontend.
- Created reusable UI components and organized learning workflows to provide a structured and engaging user experience.
- Deployed the platform on Vercel with a production-oriented frontend architecture.

- Developed a React-based contact management application for adding, viewing, editing, and maintaining contact records.
- Implemented reusable interface components and client-side logic to support contact creation and management workflows.
- Published the application through Netlify.

CERTIFICATIONS

- Agile Methodology Virtual Experience Program — Cognizant (Oct 2022)
- Python Programming Online Training — APSSDC (May–Jun 2021)

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur